ONE FUNCTION, EVERY FEATURE

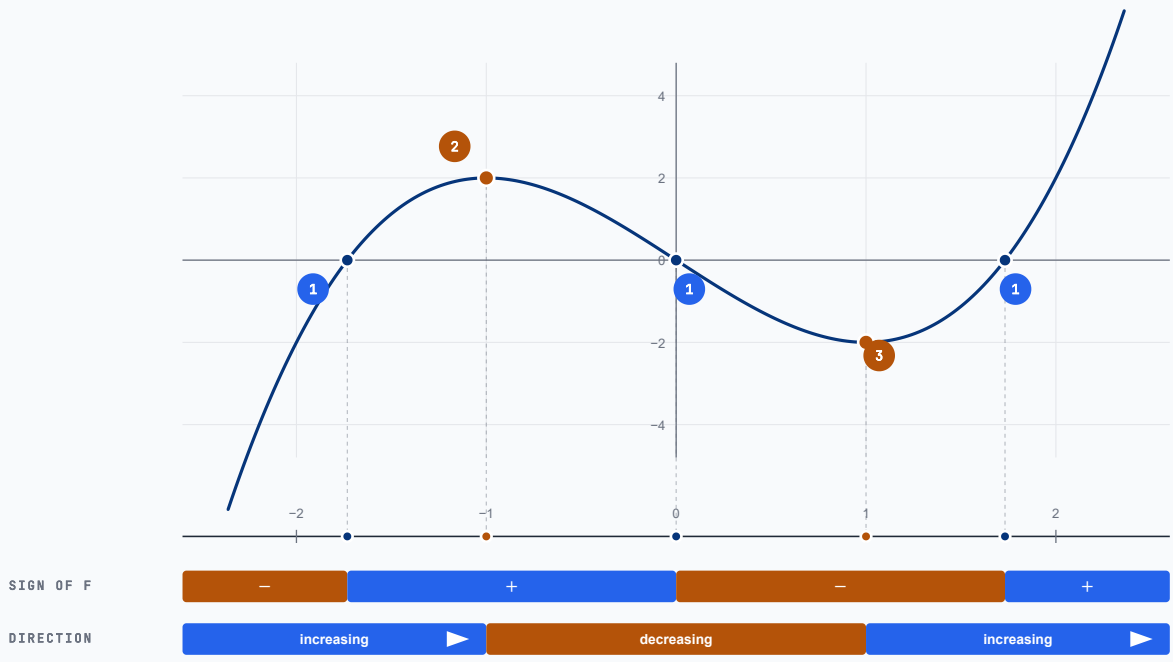
The profile of $f(x) = x^3 - 3x$

$$f(x) = x^3 - 3x = x(x^2 - 3)$$

3	2	odd
X-INTERCEPTS	LOCAL EXTREMA	SYMMETRY

01 The graph, with the sign of f and its direction along x

Strips share the graph x-axis exactly



Defined · Positive · increasing · concave up

Negative · decreasing · concave down · Numbered feature — see the table below

02 The profile, feature by feature

numbered rows match the badges on the graph

A Domain and range

	DOMAIN	$(-\infty, \infty)$ a polynomial accepts every real input
	RANGE	$(-\infty, \infty)$ the two arms run off without bound

B Intercepts

1	X-INTERCEPTS	$x = -\sqrt{3}, 0, \sqrt{3}$ from $x(x^2 - 3) = 0$
1	Y-INTERCEPT	$(0, 0)$

C Increase, decrease, extrema

2	LOCAL MAXIMUM	$(-1, 2)$
3	LOCAL MINIMUM	$(1, -2)$
	INCREASING ON	$(-\infty, -1)$ **and** $(1, \infty)$
	DECREASING ON	$(-1, 1)$

D Symmetry, periodicity, end behaviour

	SYMMETRY	**odd:** $f(-x) = -f(x)$ the graph lands on itself after a half turn about the origin
	PERIODICITY	*none*
	END BEHAVIOUR	$f(x) \to -\infty$ **as** $x \to -\infty$**;** $f(x) \to +\infty$ **as** $x \to +\infty$
	DISCONTINUITIES	*none*